

Agentic AI Connector for UHC Prior Authorization API

Design & Thought Process

1. Objective:

Design an agentic AI connector that integrates with the UnitedHealthcare (UHC) Prior Authorization API to automate, assist, and monitor prior authorization workflows in a secure, explainable, and compliant manner.

The system aims to:

- Reduce manual effort for providers
- Improve prior authorization turnaround time
- Ensure data completeness before submission
- Provide transparent explanations for approval/denial outcomes
- Maintain HIPAA-aligned handling of PHI

2. Healthcare Context: Prior Authorization:

Prior authorization is a payer-controlled process where approval is required before specific medical services, procedures, or medications are rendered.

Typical inputs:

- Patient demographics
- Provider details
- Diagnosis codes (ICD-10)
- Procedure codes (CPT/HCPCS)
- Clinical justification / notes
- Supporting documents
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Typical outcomes:

- Approved
- Denied (with reason codes)
- Pending / Additional information required

3. High-Level System Architecture:

Agentic AI Orchestrator

- └─ Validation Agent
- └─ Document Processing Agent
- └─ Planner Agent
- └─ UHC API Agent
- └─ Monitoring Agent
- └─ Explanation Agent
- └─ Dashboard & Insights Agent

4. Agent Roles & Responsibilities:

4.1 Validation Agent (Data Completeness & Compliance):

Purpose:

- Ensure all required fields are present
- Validate medical code formats
- Detect missing or inconsistent data before API

Responsibilities:

- Check mandatory UHC fields
- Validate ICD/CPT codes
- Flag missing clinical justification
- Prevent invalid API calls

Sample Agent Prompt:

You are a healthcare data validation agent.

Validate the prior authorization request for completeness and correctness.

Return:

- Missing fields (if any)
- Invalid medical codes
- Compliance risks

Do not submit the request if critical data is missing.

4.2 Planner Agent (Decision-Making Brain):

Purpose:

- Decide the next best action in the workflow

Responsibilities:

- Decide whether to:
 - Request more data
 - Submit to UHC
 - Re-submit after denial
 - Escalate to human review
- Coordinate between agents

Sample Agent Prompt:

You are a healthcare workflow planner agent.

Based on the current authorization state, decide the next action.

Possible actions:

- Request additional provider input
- Submit prior authorization
- Check authorization status
- Escalate to human review

Explain your decision briefly.

4.3 UHC API Agent (Connector Layer)

Purpose:

- Handle secure interaction with UHC Prior Authorization API

Responsibilities:

- OAuth2 / token-based authentication
- Submit authorization requests
- Poll authorization status
- Fetch decision details

Assumed Endpoints:

- POST /prior-authorization
- GET /prior-authorization/{id}/status

Security Considerations:

- Encrypted secrets
- No PHI stored in logs
- Token rotation

4.4 Monitoring Agent (Status & Retry Logic):

Purpose:

- Monitor authorization lifecycle

Responsibilities:

- Periodic status checks
- Detect stalled or pending cases
- Trigger retries or escalation
- Track SLA metrics

Sample Agent Prompt:

You are a monitoring agent for prior authorization workflows.

Track authorization status changes.

If pending exceeds threshold, recommend escalation.

4.5 Explanation Agent (Human-Readable Output):

Purpose:

- Translate payer responses into understandable language

Responsibilities:

- Convert UHC decision codes into plain English
- Summarize approval/denial reasons
- Assist provider decision-making

Sample Agent Prompt:

You are a healthcare explanation agent.

Explain the prior authorization decision in simple, non-technical language for clinicians and administrative staff.

4.6 Document Processing Agent (Clinical Attachments & Evidence):

Purpose:

- Handle supporting clinical documents required for prior authorization.
- Ensuring that all necessary attachments are present, readable, and contextually relevant before submission to UHC.

Responsibilities:

- Ingest uploaded documents (PDFs, scanned notes, lab reports)
- Extract key clinical information (diagnoses, procedures, physician notes)
- Verify required document types based on CPT/ICD codes
- Flag missing, unreadable, or irrelevant attachments
- Provide structured summaries to downstream agents
- Prevent submission if mandatory documentation is missing

Sample Agent Prompt:

You are a healthcare document processing agent.

Analyze the attached clinical documents for a prior authorization request.

Tasks:

- Identify document type (clinical note, lab report, imaging, referral)
- Extract relevant medical evidence
- Check whether required documents are missing based on the procedure
- Return a structured summary and any missing documentation alerts

Do not infer diagnoses or provide medical opinions.

4.7 Dashboard & Insights Agent (Operational Visibility):

Purpose:

Provide real-time, de-identified operational visibility into prior authorization workflows for providers and administrative teams without exposing protected health information.

Responsibilities:

- Aggregate authorization status metrics across workflows.
- Monitor turnaround time and SLA adherence
- Identify operational bottlenecks and denial trends
- Surface escalation alerts for stalled or high-risk cases
- Support audit and compliance reporting using metadata only

Sample Agent Prompt:

You are a healthcare operations dashboard agent.

Generate real-time insights from prior authorization workflows.

Tasks:

- Summarize authorization statuses and trends
- Identify delays, SLA risks, and high-denial procedures
- Present insights in a clear, non-clinical format

Do not display protected health information.

5. Agentic Workflow (End-to-End):

1. Provider submits prior auth request
2. Validation Agent checks completeness
3. If incomplete → request missing info
4. Planner Agent approves submission
5. UHC API Agent submits request
6. Monitoring Agent tracks status
7. On decision:
 - Explanation Agent generates summary
8. If denied:
 - Planner decides retry vs escalation

6. Pseudocode (Simplified):

```
def prior_auth_workflow(request):
    validation_result = validation_agent.validate(request)

    if not validation_result.is_valid:
        return validation_result.feedback

    plan = planner_agent.decide("submit_prior_auth")

    if plan.action == "submit":
        auth_id = uhc_api_agent.submit(request)
```

```
while True:
    status = uhc_api_agent.check_status(auth_id)

    if status in ["approved", "denied"]:
        explanation = explanation_agent.explain(status)
        return explanation

    if monitoring_agent.should_escalate(status):
        return "Escalated to human review"

    sleep(POLL_INTERVAL)
```

7. Compliance & Security Considerations

- HIPAA-aware handling of PHI
- No patient data used for model training
- Role-based access control
- Audit logs without PHI
- Human-in-the-loop for edge cases

8. Tech Stack (Proposed):

- **Backend:** Python, FastAPI
- **Agent Framework:** LangGraph / CrewAI-style orchestration
- **LLM:** Secure, private deployment
- **Auth:** OAuth2
- **Storage:** Encrypted database for metadata only
- **Observability:** Structured logging & metrics

9. Value Proposition:

This agentic AI connector:

- Reduces manual administrative burden
- Improves submission accuracy
- Shortens authorization cycles
- Enhances transparency and trust
- Scales across payers beyond UHC

10. Conclusion:

This design demonstrates how agentic AI can be safely and effectively applied to healthcare payer workflows. By combining structured agent roles, secure API integration, and explainable outputs, the system supports clinicians and administrators while maintaining compliance and reliability.